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Coordinated sizing of battery and charging systems for underground mining electric trucks

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A B S T R A C T

The heavy reliance on diesel-powered haul trucks in underground mines contributes substantially to CO_2 emissions and poses significant health risks to workers. Electrification of haul trucks offers a promising way to address these challenges, yet designing charging system in this constrained environment involves complex trade-offs between cost, productivity, and technical feasibility. This paper presents a novel optimisation-based framework for the coordinated sizing of onboard batteries and charging systems in underground operations, explicitly considering the interactions between battery size, payload capacity, charge rate, battery degradation, and regenerative braking. Three charging technologies, including fast charging, battery swapping, and trolley-assist, are systematically compared using tailored design models through rigorous net-present cost analysis over the mine's operational life. The framework assesses how essential operational elements, such as the minimum viable battery capacity, frequency of charging cycles, length of the trolley, and motor efficiency, impact cost and productivity results. Simulation results for a hypothetical mine reveal distinct Pareto-optimal frontiers for each charging technology, highlighting how optimal choices shift under varying technical and economic conditions. This framework provides mining planners with a quantitative basis for selecting charging strategies that balance capital and operating costs with sustained productivity.

1. Introduction

1.1. Background and motivation

In Australia, the mining industry accounts for about 10% of the total energy use in the country (both for vehicles and machinery); its main energy supply is diesel (41%) [1], which is used mainly in haul trucks. In underground mines, these vehicles transport tens of tonnes of ore or waste material to different locations underground or on the surface. As such, the electrification of the mining transport fleet has recently attracted attention as the first step towards decarbonising underground mining [2]. In addition to reducing greenhouse gas emissions, the electrification of haul trucks offers multiple operational benefits, including removing the exposure of workers to diesel fumes, substantial reductions in ventilation energy requirements, and lower operating costs [3,4].

Compared to open-pit mine haul trucks, underground haul trucks typically have smaller payload capacities, reducing the required onboard battery capacity and making electrification more technically feasible. Consequently, underground transport has become a primary driver of electrification in the mining sector. Detailed insights into the potential and advancements in mine electrification can be found in [2,3,5]. Although beneficial, the transition to electric haul trucks in underground mines encounters substantial technical and operational hurdles, especially with respect to charging infrastructure design, battery size, and alignment of charging strategies with production targets. Addressing these challenges requires systematic modelling and optimisation to identify cost-effective and operationally viable solutions.

1.2. Related studies

Extensive research has been undertaken on charging strategies and infrastructure planning for light-duty electric vehicles and public transit systems [6–8]. In the bus transport domain, for example, studies have examined optimal charging station sizing and placement, battery capacity, fleet sizing, and scheduling strategies [9–13]. While these works offer valuable methodological insights, they are predominantly designed for structured transit networks and depot-based operations. These contexts, often found in urban or highway settings, usually have

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Nomenclature

Sets and indices

i	Index of haul profile segments
j	Index of the number of battery replacements
k	Index of charging systems

Parameters

$C^{\mathrm{su,tr,sw,oc}}$	Cost of electric vehicle supply unit, transformer, swapping robot, and overhead catenary (\$/kW)
β_1/β_2	Battery degradation parameters
δ_i	Incline angle of segment i
$\eta^{\mathrm{ac/dc/tr}}$	Efficiencies of the AC/DC converter/DC-DC converter/transformer
λ^{el}	Electricity price (\$/kWh)
μ_i^{r}	Rolling resistance of segment i
$\overline{P}^{\mathrm{dch}}$	Maximum discharge power of the battery (kW)
ρ^b	Battery energy density (Wh/kg)
$C^{\mathrm{ac/dc}}$	AC/DC converter cost (\$/kW)
C^{bat}	Cost of the battery (\$/kWh)
C^{rate}	Charge/discharge rate of the battery
$C^{rp,bat}$	Used battery re-purposing cost (\$)
C_k^{fix}	Fixed cost of construction and development
E^{cell}	Energy capacity of each battery cell
EoL^b	Battery's EoL at the end of the second life
EoL^{ev}	Battery's EoL for the electric vehicle application
g	Gravitational constant
int	Interest rate
L_i	Length of segment i (m)
L_i^{st}	Starting location of the trolley line (m)
L_i^{ta}	Length of the trolley line (m)
M^{full}	Mass of fully loaded truck (kg)
M^{nb}	Mass of unloaded electric truck w/o battery (kg)
N^{bat}	Number of batteries used in operation
N^{seg}	Number of haul route segments
N^{trip}	Number of trips between consecutive charging
S_i	Truck speed in segment i
T	Operational life of the mine (years)
$T^{\mathrm{ac/dc}}$	Lifetime of the AC/DC converter
T^{op}	Total operational time per shift (hour)
$T^{\mathrm{su/tr/sw/oc}}$	Lifetime of electric vehicle supply unit, transformer, swapping robot, overhead catenary
T^{swap}	Typical swapping time
T^{trip}	Time required to complete a single trip (hour)
W^u	Used battery discount factor

Variables

$\kappa_{k,i}^b$	Health of batteries
$\omega_{k,j}^h$	Battery's health factor
$c_k^{\mathrm{ca/o/m}}$	Capital cost/operating cost/maintenance cost (\$)
c_k^{el}	Cost of electric power (\$/kWh)
c_k^{eq}	Charging infrastructure equipment cost (\$)
c_k^{re}	Battery replacement cost (\$)
dod_k	Depth of discharge in each charging cycle (%)
e^{cap}	Battery capacity (kWh)
$e^{\mathrm{str,EoL}}$	Energy storage required at the end of life of the battery (kWh)
e^{str}	Required energy storage capacity (kWh)
$e_i^{\mathrm{bt/ot/at}}$	Energy required for the truck's drivetrain before, during, and after trolley connection (kWh)
e_i^{b}	Battery energy at the start of a segment (kWh)
e_i^{seg}	Energy for traversing a segment i (kWh)
e_k^{d}	Daily electricity consumption (kWh)
$m^{\mathrm{e\text{-}truck}}$	Unloaded weight of the electric truck (kg)
$m^{\mathrm{pl,shift}}$	Payload transported per shift (kg)
m^{pl}	Payload capacity of the truck (kg)
$n^{\mathrm{full,cyc}}$	Number of full battery charging cycles per year
n^{shift}	Number of trips per shift
$n_k^{\mathrm{cyc,yr}}$	Number of charging cycles per year
$p^{\mathrm{bs,ins}}$	Installed power capacity of the battery swapping system (kW)
$p^{\mathrm{fc,ins}}$	Installed power capacity of the fast charging system (kW)
$p_i^{\mathrm{ta,ins}}$	Installed power capacity of the trolley (kW)
p_i^{ta}	Power supplied by the trolley (kW)
p_i^{dr}	Power of the electric truck drivetrain in segment (kW) i
t^{ch}	Time required for battery charging (hour)
t^{swap}	Time required for battery swapping (hour)
$v^{\mathrm{ac/dc}}$	AC/DC converter salvage value (\$)
v^{bat}	Battery salvage value (\$)
v^{su}	Electric vehicle supply unit salvage value (\$)
v^{tr}	Power transformer salvage value (\$)
v_{bs}^{sw}	Swapping robot salvage value (\$)
v_k	Salvage value (\$)
v_{ta}^{oc}	Trolley line salvage value (\$)

limited physical constraints and predictable conditions. In contrast, underground mining constitutes an exceptionally challenging application environment, characterised by confined spaces, stringent safety requirements, and productivity-critical operations. In this setting, the design and deployment of effective charging systems for electric haul trucks present unique technical and operational complexities.

The mining industry's desire to transition to electric trucks occurs at a time of technological readiness, due to the shift of domestic transportation to electric in recent years. As a result, original equipment manufacturers (OEMs) have already released a range of electric mining trucks. A prominent example is the Sandvik TH665B (one of the largest underground electric trucks with a payload capacity of 65 tonnes), which was recently unveiled and successfully tested at the Sunrise Dam gold mine in Western Australia [14]. However, substantial research and development efforts are still required to address key issues in terms of timing, location, charging systems, and power levels to charge electric mining fleets, as highlighted in recent industry reports by the State of Play [15] and the Electric Mine Consortium [16,17].

Despite this momentum, there remains a notable gap in academic research on the integrated design of charging systems and battery sizing tailored specifically to underground haulage operations. As such, existing studies on underground electric transport are sparse. Ref. [18] investigated fast charging and battery swapping for underground trucking, but the analysis does not provide a coordinated design framework that jointly sizes the onboard battery and charging system.

In this domain, [19] proposed an iterative method to minimise life cycle cost by optimising battery size, taking into account specific truck details and fleet size. Although this advances design realism, this research still lacks integration of charging system design with battery sizing and does not address trolley-assist systems as an emerging solution for heavy-duty mining trucks. Using a 3D discrete event simulation of an open-pit mine, [20] evaluated three battery-electric haulage configurations to compare their operational performance, energy efficiency, and productivity based on current industry technologies and readiness. However, the study does not incorporate a strategic charging system design coordinated with battery sizing, nor does it include a cost–benefit analysis.

Trolley-assist charging systems have been investigated in studies such as [21,22]. The former introduces a simulation tool to assess the productivity and energy consumption of electric haul trucks in an open-pit mine, while the latter evaluates the life cycle costs of trolley-based electric trucks in heavy-duty applications. More recent studies have continued to advance trolley-assist haulage design. Ref. [23] examined the energy consumption and required onboard battery capacity for battery–trolley electric trucks in surface mines, highlighting how trolley coverage and haul-road geometry affect energy demand and battery sizing. Ref. [24] proposed a techno-economic sizing framework for trolley-assist battery-electric haul trucks, jointly determining onboard battery capacity and e-road length through a surrogate-model optimiser coupled with dynamic simulation and a battery degradation model. While these studies offer valuable insights, they are limited to surface operations and consider only a single trolley–battery powertrain configuration. They do not address the operational characteristics unique to underground haulage, including fixed haul profiles, limited feasible charging locations, and the strong trade-offs between battery mass and productivity.

Overall, the existing literature either focuses on a single charging technology, addresses battery sizing or charging infrastructure in isolation, or is limited to surface mining contexts. None of the above studies provides a scenario-based design with technology-specific optimisation that jointly evaluates fast charging, battery swapping, and trolley-assist systems for underground haulage, while accounting for battery degradation, charging system sizing, and productivity impacts within an integrated techno-economic model. Addressing this gap forms the central contribution of the present study.

1.3. Novelties and contributions

This paper presents a novel framework for the design of a cost-effective underground mining charging system with specific results for a typical underground haul profile. The framework incorporates fast charging, battery swapping, and overhead trolley-assist technologies and captures their interactions with onboard battery capacity, truck speed, and mining productivity. For each charging system, the required onboard battery capacity, lifetime cost, and productivity implications are evaluated. The key contributions of this work are:

- Coordinated sizing of the power rating for charging systems and the capacity of onboard batteries
- Development of nonlinear programming (NLP) optimisation models for fast-charging and battery swapping systems design
- Development of a linear programming (LP) optimisation formulation for designing a trolley assist charging system
- Comparative assessment of different charging solutions considering battery degradation and replacement throughout the mine life in a comprehensive net-present cost (NPC) framework

In conclusion, from an energy systems perspective, the proposed framework contributes to the planning and techno-economic evaluation of high-power charging infrastructure for large electric loads in constrained environments.

This paper is organised as follows. Section 2 outlines the problem statement. Section 3 explains the design process for various charging systems. Section 4 presents the cost analysis framework. Section 5 discusses the simulation results. Finally, Section 6 concludes the paper and suggests directions for future research.

2. Problem statement

The design of the charging system for electric mining haul trucks represents a multifaceted decision-making challenge characterised by conflicting goals and strict constraints. For example, a larger onboard battery reduces the need for frequent recharging, probably enhancing productivity, but its increased weight might reduce the payload capacity. Similarly, selecting a specific battery system introduces various costs, charging rates, and energy density, which simultaneously exerts positive and negative effects on productivity and cost. As such, the ultimate goal is to identify the optimal type, size, and location of various components, with the aim of minimising both capital expenditure (CAPEX) and operational expenditure (OPEX) while maximising productivity.

The design problem involves numerous nonlinear and non-convex relationships, making the formulation and solution of a comprehensive optimisation model highly complex. To address this, a scenario-based design approach is adopted, in which an optimisation model is independently applied to each charging system across a range of representative scenarios. This enables a structured exploration of key trade-offs, such as cost, energy storage requirements, and operational productivity, and supports the construction of a Pareto-optimal front to guide informed and balanced decision-making.

Fig. 1 presents the high-level block diagram of the proposed framework. The model incorporates a wide range of inputs, including economic data (e.g., cost models for each charging technology and truck battery), mining-specific data (e.g., haul profiles and equipment specifications), and other technical data (e.g., charging system characteristics and operational constraints). Various scenarios are then defined based on the type of charging technology, truck speed, charging location, charging/swapping frequency, and length of the trolley system. For each scenario, a hauling simulation is performed to determine the required onboard battery capacity, the charging system power rating, and the achievable truck payload (i.e., the material transported in a shift). The framework ultimately evaluates the cost–payload trade-offs for each charging system and scenario, producing a Pareto-optimal set of solutions to guide balanced and evidence-based investment decisions.

This study focuses exclusively on the design of the electrical charging system for underground mining operations that is required to support battery-electric haul trucks, where all traction and charging processes are powered by electricity. It does not cover the generation of the electrical energy, which for remote mines would normally be from fossil fuels and possibly supplemented by renewable energy sources. Multi-energy couplings (e.g., heat or gas) are out of the scope of this study, as the objective is to design the charging infrastructure rather than model energy supply sources.

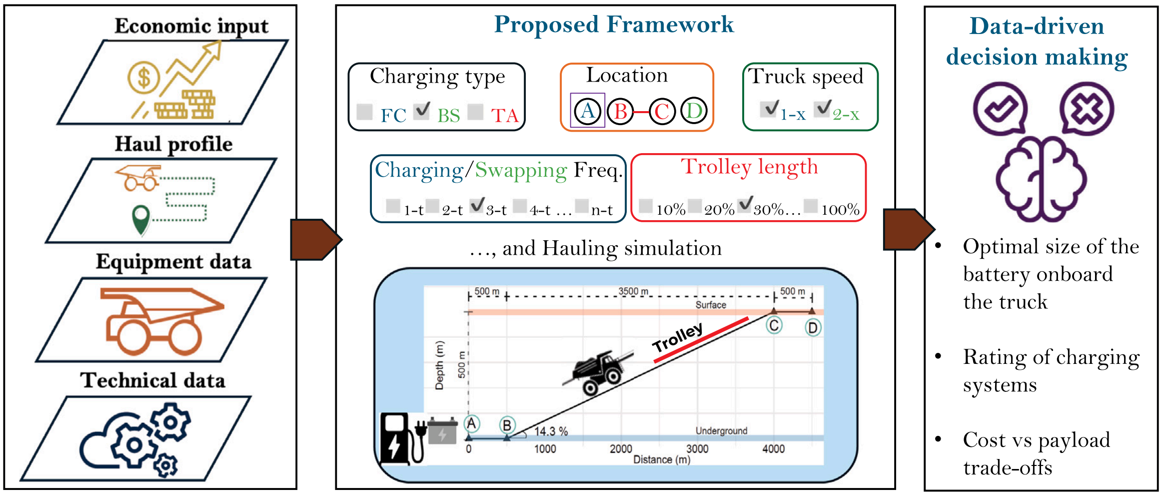

Fig. 1. High-level block-diagram of the proposed framework, including the input and output parameters.

3. Methodology and design framework

This section presents the mathematical frameworks developed to generate design parameters for each charging system.

To do so, first, we need to estimate the electrical power and energy requirements for the haul trips. The energy required consists of three components: (1) height change (potential energy), (2) rolling resistance, and (3) air resistance. Of the three factors, the energy associated with air resistance is typically disregarded because of the low maximum speed of a haul truck operating in an underground mine. The speed of the haul truck in different sections of a haul profile affects the required battery power for the drivetrain. The maximum power required from the battery may also affect the required energy capacity of the battery. This is because the maximum charge and discharge power of a battery is limited by its C-rating. For modelling purposes, we followed standard textbook methods to formulate the energy and power requirements [25].

There are several key parameters that can have a substantial impact on the design, which are explained below.

1. Increasing the battery capacity reduces the charging frequency, but also increases the weight of the unloaded electric truck, and hence reduces its payload capacity.
2. The C-rating of the battery could be a constraining factor when high peak power is required, leading to an oversized battery.
3. The higher the uphill speed of the truck, the greater the productivity, but also the higher the power demand, which increases the motor size and possibly the battery size.
4. The battery charging frequency for fast charging and swapping systems and the length of the trolley assist affect the size of the onboard battery, the productivity, and the required capacity of the charger.

In addition, throughout the design procedure, battery operation in deep-discharge conditions is minimised, which is achieved by a conservative minimum state of energy limit together with a C-rate based discharge power bound. This ensures that fully loaded uphill segments occur within a mid to high-SOC window where the battery can deliver its rated power, so that the model does not rely on operation in low-SOC regions where discharge power capability would be degraded.

In the following subsections, we will explain the mathematical models for the three different charging systems.

3.1. Fast charging system

In contrast to a trolley-assist system, electric haul trucks operating with fast charging or battery swapping systems must complete a predefined number of haul cycles before accessing the charging infrastructure. To maximise operational productivity under such constraints, a nonlinear programming (NLP) framework that co-optimises truck payload capacity, battery size, and fast charging system power is developed. The objective of the model is to maximise the total transported payload per shift while ensuring compliance with key constraints, including vehicle mass limits, battery C-rate limits, and operational shift duration, which are given in (1a)–(1m).

$$\underset{e^{\mathrm{cap}},\, p^{\mathrm{fc,ins}}}{\text{maximize}} \quad m^{\mathrm{pl,shift}} - \alpha \cdot p^{\mathrm{fc,ins}} \tag{1a}$$

$$\text{subject to} \quad m^{\mathrm{pl,shift}} = n^{\mathrm{shift}} \cdot m^{\mathrm{pl}} \tag{1b}$$

$$m^{\mathrm{pl}} = M^{\mathrm{full}} - m^{\text{e-truck}} \tag{1c}$$

$$m^{\text{e-truck}} = M^{\mathrm{nb}} + \frac{e^{\mathrm{cap}}}{\rho^{\mathrm{b}}} \tag{1d}$$

$$m_i = M^{\mathrm{full}}, \quad \forall i \leq |I|/2 \tag{1e}$$

$$m_i = m^{\text{e-truck}}, \quad \forall i > |I|/2 \tag{1f}$$

$$e^{\mathrm{cap}} = \max\left(\frac{e^{\mathrm{str}}}{0.8}, \frac{\overline{P}^{\mathrm{dch}}}{C^{\mathrm{rate}}}\right) \tag{1g}$$

$$e^{\mathrm{str}} = \max_{i} \sum_{i=1}^{I} e_i^{\mathrm{seg}}, \quad \forall I \in \{1, 2, \ldots, N^{\mathrm{trip}} \cdot N^{\mathrm{seg}}\} \tag{1h}$$

$$n^{\mathrm{shift}} = \frac{T^{\mathrm{op}}}{T^{\mathrm{trip}} + \frac{t^{\mathrm{ch}}}{N^{\mathrm{trip}}}} \tag{1i}$$

$$t^{\mathrm{ch}} = \frac{e^{\mathrm{trip}}}{p^{\mathrm{fc,ins}}} \tag{1j}$$

$$p^{\mathrm{fc,ins}} \leq C^{\mathrm{rate}} \cdot e^{\mathrm{cap}} \tag{1k}$$

$$e^{\mathrm{trip}} = \sum_{i=1}^{N^{\mathrm{trip}} \cdot N^{\mathrm{seg}}} e_i^{\mathrm{seg}} \tag{1l}$$

$$e_i^{\mathrm{seg}} = m_i \cdot g \cdot L_i \cdot \left(\mu_i^{\mathrm{r}} + \sin \delta_i\right) \tag{1m}$$

In practice, high-capacity lithium-ion batteries cannot sustain their maximum charging power across the full SOC range. To ensure a realistic system-level representation, this study adopts a moderate 2C (average) value of the fast-charging rate, consistent with the capability of emerging high-power industrial batteries for underground mining trucks. Furthermore, the useable battery capacity is limited to 80%

of its nominal value, reflecting end of life degradation and avoiding operation in the high SOC saturation region where charging power naturally reduces. This modelling choice provide a conservative and industry-aligned approximation of fast-charging behaviour.

The objective function (1a) maximises productivity in a working shift and penalises excessive charging power rating by reflecting its capital expenditure in the objective function. Eq. (1b) links the payload in a working shift to the number of haul cycles, which depends on the number of haul trips the electric truck is able to complete and the charging duration. After dumping the load, the electric truck is assumed to return via the same route, completing a round trip. As a result, the weight of the truck along different segments depends on the battery size and its loading condition. Consequently, the payload per cycle is limited by the mass composition of the truck, as defined in Eqs. (1c)–(1f), where the electric components include both the battery mass and the non-battery mass (e.g. electric motor and power electronics). Eq. (1g) ensures that battery capacity satisfies both energy and power demands, considering battery degradation to 80% of its initial capacity at the end of its life. The energy storage required in the most energy-intensive set of haul segments is captured by (1h). The number of feasible haul cycles per shift is calculated in (1i), accounted for the travel time per trip and the average charging time per trip. Eqs. (1j) and (1k) define the charging time and enforce the C-rate limit on the fast charging power. The total energy consumed before charging is derived in (1l), based on the segment-level energy demand (1m), which includes the rolling resistance and gradient effects. Overall, this framework ensures a feasible and economically viable design for fast charging systems in mining haulage applications.

In addition to operational performance and power requirements, it is important to evaluate long-term battery usage metrics that impact cost and system sustainability, including the number of equivalent full-charging cycles per year ($n^{\text{full,cyc}}$). This is approximately given by (2).

$$n^{\text{full,cyc}} \approx \frac{\sum_{i=1}^{n^{\text{shift}} \times N^{\text{seg}}} e_i^{seg}}{e^{\text{cap}} N^{\text{bat}}} \times 2\,\frac{\text{shifts}}{\text{day}} \times 365\,\frac{\text{days}}{\text{year}} \tag{2}$$

where $N^{\text{bat}} = 1$ for fast charging and trolley-assist, while for battery swapping, $N^{\text{bat}} = 2$, indicating that one battery remains in the truck as another charges.

Furthermore, since each scenario yields a different payload and overall cost, making fair comparisons is challenging. A method to address this is by calculating the cost per tonne of the transported material, using the index presented in (3).

$$AC2L_k = \frac{AC_k^{\text{tot}}}{365 \times 2 \times m_k^{\text{pl,sh}}}, \ \forall k \tag{3}$$

$$AC_k^{\text{tot}} = \frac{int(1+int)^T}{(1+int)^T - 1} C^{\text{tot}}, \ \forall k \tag{4}$$

In (3), the numerator is the total annual cost, calculated by (4), divided by the annual amount of the transported ore body. Although the daily extraction of mining materials is not constant during a year and depends on various factors (e.g., season and commodity price), the AC2L index remains an acceptable metric because the amount of extraction directly affects the operation and maintenance cost and this cost is a major part of the NPC. Therefore, the AC2L metric represents the annualised cost per tonne of material hauled, providing a normalised cost-per-tonne basis for comparing the different charging strategies.

3.2. Battery swapping system

The battery swapping system benefits from reduced charging power requirements and less operational delays, which can enhance overall productivity. However, this approach necessitates the availability of spare batteries and dedicated swapping infrastructure, potentially increasing the system's capital costs.

In contrast to the fast charging framework, formulating a mathematical optimisation framework for the battery swapping system requires adjustments to (1i), (1j), and (1k), which are given below.

$$\text{maximize} \quad m^{\text{pl,shift}} - \alpha \cdot p^{\text{bs,ins}} \tag{5a}$$

$$\text{subject to} \quad (1b)\text{–}(1h) \tag{5b}$$

$$n^{\text{shift}} = \frac{T^{\text{op}}}{T^{\text{trip}} + \frac{t^{\text{swap}}}{N^{\text{trip}}}} \tag{5c}$$

$$t^{\text{swap}} = \max\left(T^{swap}, m^{\text{bat}} \times \frac{T^{swap}}{M^{swap}}\right) \tag{5d}$$

$$p^{\text{bs,ins}} = \frac{e^{\text{trip}}}{T^{\text{trip}} \cdot N^{\text{trip}}} \tag{5e}$$

$$(1l)\text{–}(1m) \tag{5f}$$

The electric truck completes a number of haul trips and then swaps its battery for a fully charged one. Therefore, the total number of haul cycles per shift is calculated by (5c), which is a function of the swapping time. In addition, the swapping time is estimated as a function of the battery mass, based on the swapping time of the Sandvik TH550B battery [26], as in (5d). Since at least two batteries are required, the depleted battery can be slowly charged while the truck operates with the other battery. This reduces the required charging power compared to fast charging, presented by (5e).

3.3. Overhead trolley assist system

Employing overhead trolley lines enables trucks to receive power while in motion, removing the necessity to halt for fast charging or battery swapping; consequently, this enhances productivity. However, trolley lines require significant capital investment and offer limited flexibility for future mining development. So, it is desirable to use them as long as possible during the mine's life and to minimise their length. Intuitively, the most advantageous location for installing trolley lines is on the slope where the highest power and energy are required. In this way, trolley lines can provide power to go uphill and, at the same time, recharge the batteries if needed. An alternative operating approach is to simultaneously supply the necessary power for the electric truck's drivetrain going uphill from both the trolley system and the truck battery. In this case, careful design is needed to find the optimal combination of battery capacity and maximum trolley charging power. During downhill travel, the trolley system can partially or fully recharge the battery by supplementing the energy from regenerative braking, or it might opt not to recharge the battery. Thus, by accounting for the time the trolley system is engaged in both uphill and downhill states (determined by the trolley's length and the speed of the electric truck), along with the battery's maximum charging power, a balance must be struck between the battery capacity and the power capacity of the trolley system.

To formulate the coordinated design of the onboard battery for the electric truck and the power rating of the trolley system, three principles should be taken into account.

1. The state of energy of the onboard battery at each point in the haul profile should remain within predefined safe bounds.
2. In the general case, the haul profile segment designated for the trolley line, such as the ramp, can be divided into three subsegments due to the positioning of the overhead trolley line, as illustrated in Fig. 2.
3. When connected to the trolley line, the energy required for the truck drivetrain can come from the battery and/or trolley system.

As shown in Fig. 2, in the first subsegment (B–B′), the truck uses the battery onboard to reach the trolley line at the point B' when going uphill. Hence, the battery state of energy limitation must be enforced

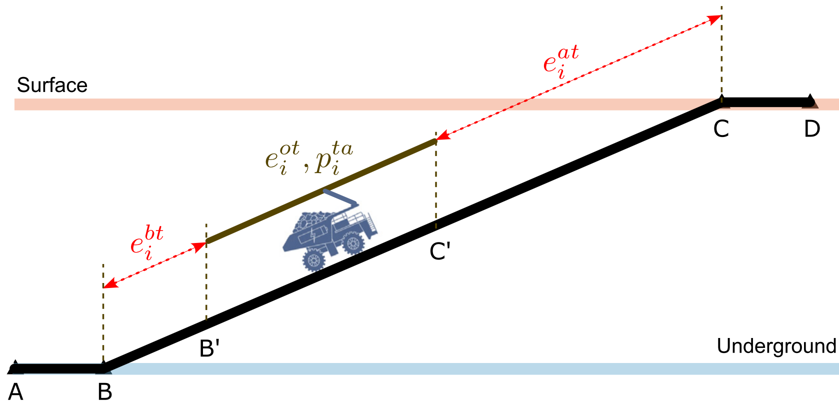

Fig. 2. Conceptual illustration for trolley system energy requirement.

in the optimisation problem. However, in the downhill trip, the battery could be charged by regenerative braking. Therefore, battery charging constraints should also be considered for this subsegment.

Since the drivetrain power could be supplied by the onboard battery, the trolley system, or both when connected to the trolley line, there must be two constraints to avoid overcharging or running out of charge while the truck is connected to the trolley line. Once detached from the trolley line at point C, the truck must draw the necessary energy for its drivetrain from the onboard battery when climbing the ramp. During downhill trips, the battery could be charged by regenerative braking; therefore, overcharging should be avoided. Accordingly, six inequality energy constraints should be considered for each segment of the haulage profile. Note that some of these constraints overlap; hence, unnecessary constraints are removed to avoid duplication.

In (6a)–(6u), a linear optimisation model is proposed that takes into account all these design challenges and finds the optimal balance between battery capacity and power level in the trolley system. Because the trolley location, its length, and the truck speed are scenario-defined rather than decision variables, the segment-level energy terms ((6g)–(6k)) remain affine. As a result, the linear programming formulation is exact and does not rely on any relaxation or approximations.

$$\max\ m^{\text{pl,shift}} - \alpha \cdot \sum_i p_i^{\text{ta,ins}} \tag{6a}$$

s.t.

$$e_i^{\text{b}} - e_i^{\text{bt}} \geq 0, \quad \forall i \tag{6b}$$

$$e_i^{\text{b}} - e_i^{\text{bt}} - e_i^{\text{ot}} + e_i^{\text{ta}} \leq e^{\text{cap}} EoL^{ev}, \quad \forall i \tag{6c}$$

$$e_i^{\text{b}} - e_i^{\text{bt}} - e_i^{\text{ot}} + e_i^{\text{ta}} - e_i^{\text{at}} \geq 0, \quad \forall i \tag{6d}$$

$$e_i^{\text{b}} - e_i^{\text{bt}} - e_i^{\text{ot}} + e_i^{\text{ta}} - e_i^{\text{at}} \leq e^{\text{cap}} EoL^{ev}, \forall i \tag{6e}$$

$$e_i^{\text{b}} = e_{i-1}^{\text{b}} - e_{i-1}^{\text{bt}} - e_{i-1}^{\text{ot}} + e_{i-1}^{\text{ta}} - e_{i-1}^{\text{at}}, \quad \forall i \neq 1 \tag{6f}$$

$$e_i^{\text{bt}} = m_i g L_i^{\text{st}} \left(\mu_i^r + \sin\delta_i\right), \quad \forall i \tag{6g}$$

$$e_i^{\text{ot}} = m_i g L_i^{\text{ta}} \left(\mu_i^r + \sin\delta_i\right), \quad \forall i \tag{6h}$$

$$e_i^{\text{at}} = m_i g L_i \left(\mu_i^r + \sin\delta_i\right) - e_i^{\text{ot}} - e_i^{\text{bt}}, \ \forall i \tag{6i}$$

$$p_i^{\text{dr}} = m_i g S_i \left(\mu_i^r + \sin\delta_i\right), \quad \forall i \tag{6j}$$

$$e_i^{\text{ta}} = p_i^{\text{ta}} \frac{L_i^{\text{ta}}}{S_i}, \quad \forall i \tag{6k}$$

$$m_i = M^{\text{full}}, \quad \forall i \leq I/2 \tag{6l}$$

$$m_i = M^{\text{nb}} + \frac{e^{\text{cap}}}{\rho^{\text{b}}}, \quad \forall i \geq I/2 + 1 \tag{6m}$$

$$m^{\text{pl,shift}} = n^{\text{shift}} \left(M^{\text{full}} - (M^{\text{nb}} + \frac{e^{\text{cap}}}{\rho^{\text{b}}}) \right) \tag{6n}$$

$$0 \leq p_i^{\text{ta}} \leq p_i^{\text{ta,ins}}, \quad \forall i \leq I/2 \tag{6o}$$

$$0 \leq p_i^{\text{ta}} \leq p_{I-i+1}^{\text{ta,ins}}, \quad \forall i \geq I/2 + 1 \tag{6p}$$

$$p_i^{\text{ta}} - p_i^{\text{dr}} \leq e^{\text{cap}} C^{\text{rate}}, \quad \forall i \tag{6q}$$

$$p_i^{\text{dr}} - p_i^{\text{ta}} \leq e^{\text{cap}} C^{\text{rate}}, \quad \forall i \tag{6r}$$

$$p_i^{\text{dr}} \leq e^{\text{cap}} C^{\text{rate}}, \quad \forall \{i | L_i^{\text{ta}} < L_i\} \tag{6s}$$

$$e_I^{\text{b}} - e_I^{\text{bt}} - e_I^{\text{ot}} + e_I^{\text{t}} - e_I^{\text{at}} = e_1^{\text{b}} \tag{6t}$$

$$E^{\text{min}} \leq e_i^{\text{b}} \leq e^{\text{cap}} EoL^{ev}, \quad \forall i \tag{6u}$$

The objective is to find the maximum payload per shift, as modelled by (6a). The energy constraints explained in the problem definition are enforced by (6b)–(6e). Eq. (6f) links the state of energy of the battery in different segments of the haul profile. Eqs. (6g)–(6i) model the required energy before/on/after the trolley line. Eq. (6j) represents the power for the electric truck drivetrain. Eq. (6k) establishes the relationship between the energy supplied by the trolley system and the power needed to supply that energy according to the time the truck is connected to the trolley system (L_i^{ta}/S_i). In this model, after dumping the load, the electric truck is assumed to return via the same route to complete a loop. Accordingly, the weight of the electric truck in different segments is a function of the size of the battery and the loading condition of the truck, which is expressed in (6l) and (6m). Payload per shift is defined in (6n). Unlike fast charging and battery swapping systems, the number of feasible haul cycles per shift remains constant ($\frac{T^{\text{op}}}{T^{\text{trip}}}$), as there are no charging or swapping interruptions. If there is a trolley system in segment i on the outbound trip, the same trolley system must be considered for the return trip. Since the required power can be different for the inbound and outbound trips in the same segment, the installed power of the trolley system is the maximum value of the trolley power in the two directions. This is modelled in (6o) and (6p). The charging and discharging power rate of the battery in each segment is limited by the C-rating, represented by (6q), (6r), and (6s). The model assumes that the energy consumption for subsequent cycles mirrors that of the initial cycle. For this trolley case, the steady-state energy balance is maintained by ensuring that the onboard battery's state of energy at the endpoint of the haul profile matches its state of energy at the start, as outlined in (6t). Finally, (6u) indicates that the maximum state of energy of the battery is equal to its capacity.

In addition to the battery capacity and the power level of the trolley system, the equivalent full charging cycles and AC2L metrics are estimated as presented in (2) and (3).

In all three charging system formulations, the design parameters, including battery capacity (e^{cap}), installed charging or trolley power rating ($p^{\text{fc,ins}}, p^{\text{bs,ins}}, p_i^{\text{ta,ins}}$), and resulting payload ($m^{\text{pl,shift}}$), are incorporated directly as optimisation variables. These variables appear explicitly in the mass–payload relationships, C-rate limits, charging/swapping time expressions, and the segment-level energy-balance constraints.

4. Lifetime cost evaluation framework

Cost analysis is an integral part of system planning and decision making. In this specific case, the cost analysis informs mining companies about the trade-off between productivity and the cost of implementing different charging systems and battery configurations for underground mining trucks. As a result, the most cost-effective design can be selected on the basis of productivity goals. Taking into account the high capital cost of fleet electrification and the downstream benefits (e.g., lower maintenance and operation cost) and cost (e.g., battery replacement cost), an NPC analysis is used for the duration of mine life. This section provides a comprehensive formulation of the transition costs for different charging technologies, including construction and development cost, operational cost, maintenance cost, and salvage value, as shown in (7).

$$c_k^{\mathrm{tot}} = c_k^{\mathrm{ca}} + c_k^{\mathrm{o}} + c_k^{\mathrm{m}} - v_k,\ \forall k \tag{7}$$

where $k \in \{\mathtt{fc}, \mathtt{bs}, \mathtt{ta}\}$.[1]

The design stage optimisation (Section 3) produces the key parameters used in the lifetime cost model, including the onboard battery capacity, installed charging power rating, charging energy requirement, and estimated annual battery charging cycle count. These outputs directly determine the capital, operational, replacement, and salvage cost components in the lifetime cost calculation. In addition, the payload-per-shift variable ($m^{\mathrm{pl,shift}}$) affects cost only through the normalised cost-per-ton metric (AC2L) in (3) and (4), rather than contributing to the absolute lifetime cost.

4.1. Capital cost

The capital cost for each charging system depends on the cost of materials and structural preparation, engineering design [27] and the equipment required, as in (8).

$$c_k^{\mathrm{ca}} = C_k^{\mathrm{fix}} + c_k^{\mathrm{eq}},\ \forall k \tag{8}$$

The cost of the components shown in (8) for fast charging technology is as follows.

$$c_{\mathtt{fc}}^{\mathrm{eq}} = \frac{p^{\mathtt{fc},\mathrm{ins}}}{\eta^{\mathrm{ac}}\eta^{\mathrm{dc}}\eta^{\mathrm{tr}}}C^{\mathrm{tr}} + \frac{p^{\mathtt{fc},\mathrm{ins}}}{\eta^{\mathrm{ac}}\eta^{\mathrm{dc}}}C^{\mathrm{ac/dc}} + \frac{p^{\mathtt{fc},\mathrm{ins}}}{\eta^{\mathrm{dc}}}C^{\mathrm{su}} \qquad + e_{\mathtt{fc}}^{\mathrm{cap}}C^{\mathrm{bat}} \tag{9}$$

For battery swapping infrastructure, the cost of the equipment is calculated using (10). It is important to note that in this study only one swapping robot/crane is needed because only a single electric truck operation is analysed.

$$c_{bs}^{\mathrm{eq}} = \frac{p^{\mathrm{bs,ins}}}{\eta^{\mathrm{ac}}\eta^{\mathrm{dc}}\eta^{\mathrm{tr}}}C^{\mathrm{tr}} + \frac{p^{\mathrm{bs,ins}}}{\eta^{\mathrm{ac}}\eta^{\mathrm{dc}}}C^{\mathrm{ac/dc}} + N^{\mathrm{bat}}e_{\mathtt{bs}}^{\mathrm{cap}}C^{\mathrm{bat}} + \frac{p^{\mathrm{bs,ins}}}{\eta^{\mathrm{dc}}}C^{\mathrm{su}} + C^{sw} \tag{10}$$

For overhead charging infrastructure, based on the conceptual design in [22], the estimated cost of the equipment is shown in (11).

$$c_{ta}^{\mathrm{eq}} = \frac{p^{\mathrm{ta,ins}}}{\eta^{\mathrm{ac}}\eta^{\mathrm{tr}}}C^{\mathrm{tr}} + \frac{p^{\mathrm{ta,ins}}}{\eta^{\mathrm{ac}}}C^{\mathrm{ac/dc}} + e_{\mathtt{ta}}^{\mathrm{cap}}C^{\mathrm{bat}} + L^{\mathrm{ta}}C^{\mathrm{oc}} \tag{11}$$

4.2. Operation and maintenance cost

The operating cost consists of the cost of the electric power demand and the cost of replacing the battery at the end of its useful life throughout the mine life, as shown in (12).

$$c_k^{\mathrm{o}} = c_k^{\mathrm{el}} + c_k^{\mathrm{re}},\ \forall k \tag{12}$$

The cost of electricity is a future cost that needs to be converted to the present value, as in (13), similar to other recurring costs. Typically, the cost of electricity provided to off-grid mining sites has fixed and variable components. Although variable cost could change annually depending on location, regulations, subsidies, etc., the fixed cost is typically negotiated for long-term and is generally unchanged. As a result, for simplicity, this cost is assumed constant in this study.

$$c_k^{\mathrm{el}} = \sum_{t=1}^{T}\left(365 e_k^{\mathrm{d}}\lambda^{\mathrm{el}}\left(\frac{1}{1+int}\right)^{t-1}\right),\ \forall k \tag{13}$$

The cost of battery replacement is modelled by estimating the useful life of the battery using the approach detailed in [28]. Based on the model, the lifespan of a battery depends on the charging rate, the depth of discharge of the battery, the ambient temperature, and the number of charging cycles. This comprehensive model improves accuracy in evaluating the cost competitiveness of different charging technologies and battery capacity configurations. The calculated lifetime of the batteries will determine the number of replacement batteries and when they are needed, hence the corresponding present value. In this way, (14)–(16) model the cost of battery replacement throughout the operating horizon, the battery life, and the annual percentage of loss of battery energy capacity, respectively. Despite projections that indicate a substantial decrease in battery prices in the future, we assume that the battery purchase price remains constant throughout the mine life to ensure our calculations remain reasonably conservative.

$$c_k^{\mathrm{re}} = \sum_{j=1}^{\lfloor T/t_k^{bat}\rfloor}\left(\frac{e_k^{\mathrm{cap}}C^{\mathrm{bat}}}{(1+int)^{\lfloor j\times t_k^{\mathrm{bat}}\rfloor}}\right),\ \forall k \tag{14}$$

$$t_k^{\mathrm{bat}} = \frac{1-EoL^{ev}}{e_k^{\mathrm{loss,yr}}},\ \forall k \tag{15}$$

$$e_k^{\mathrm{loss,yr}} = \beta_1 e^{\beta_2\frac{p^{\mathrm{k,ins}}}{c_k^{\mathrm{b}}}}\left(n_k^{\mathrm{cyc,yr}} \times dod_k \times E^{\mathrm{cell}}\right),\ \forall k \tag{16}$$

The only replacement cost considered in this study is for the onboard battery. Its degradation over the mine's operational lifetime (T) necessitates periodic replacement, as shown in (14)–(16). Other components, including converters, transformers, and trolley lines, are assumed to operate without needing replacement over the project horizon. Their contributions are instead captured through their respective maintenance costs and salvage values at the end of the mine's operational life.

The annual maintenance cost is considered as a fraction of the investment cost in all facilities and required equipment using the maintenance rate ρ, as shown in (17). The annual maintenance cost can change over the years of mining operation. This is represented using a linearly increasing cost based on the inflation rate.

$$c_k^{\mathrm{m}} = \sum_{t=1}^{T}\rho \times c_k^{\mathrm{ca}}\left(\frac{1+inf}{1+int}\right)^{t-1},\ \forall k \tag{17}$$

4.3. Salvage value

Upon reaching the end of their operational life, certain components of the mine retain financial value, as they can be reused for different applications or refurbished for use in other mining operations. Therefore, considering salvage values is essential for an accurate and fair cost estimate. Estimating salvage values for each charging system depends on its reusable equipment. In this way, (18) shows the salvage value of the reusable components for the fast charging infrastructure.

$$v_{fc} = v_{fc}^{\mathrm{tr}} + v_{fc}^{\mathrm{ac/dc}} + v_{fc}^{\mathrm{su}} + v_{fc}^{\mathrm{bat}} \tag{18}$$

For battery swapping infrastructure, the salvage value is also considered for swapping robots, which is expressed in (19).

$$v_{bs} = v_{bs}^{\mathrm{tr}} + v_{bs}^{\mathrm{ac/dc}} + v_{bs}^{\mathrm{su}} + v_{bs}^{\mathrm{bat}} + v_{bs}^{\mathrm{sw}} \tag{19}$$

Finally, the salvage value of trolley assist infrastructure is calculated using (20).

$$v_{ta} = v_{ta}^{\mathrm{tr}} + v_{ta}^{\mathrm{ac/dc}} + v_{ta}^{\mathrm{bat}} + v_{ta}^{\mathrm{oc}} \tag{20}$$

[1] fc is for fast charging, bs for battery swapping, and ta for trolley assist.

Salvage values are the estimated resale values at the end of the operating horizon, so they must be discounted using the future value discount factor ($DF = \frac{1}{(1+int)^{T-1}}$). Estimating the value of a second-hand component can be done in different ways. Here, the double declining balance depreciation method is used for all equipment except the battery, as shown in (21)–(25). Eqs. (26)–(27) model the total salvage values of batteries using the formulation proposed in [29]. In this model, the value of a used battery is proportional to the cost of the new battery taking into account the battery's health factor, the used battery discount factor,[2] and the re-purposing cost, given in (27). Thus, the first term in (26) is the sum of the salvage values of the batteries replaced throughout the operating horizon, and the second term is the salvage value of the last battery. Eqs. (28) and (29) calculate the value of the health factor and the health of the last battery and the others.

$$\upsilon_k^{\mathrm{tr}} = \frac{p^{\mathrm{k,ins}} C^{tr}}{\eta^{\mathrm{ac}} \eta^{\mathrm{dc}} \eta^{tr}} \left(1 - \frac{2}{T^{\mathrm{tr}}}\right)^T \times DF, \ \forall k \tag{21}$$

$$\upsilon_k^{\mathrm{ac/dc}} = \frac{p^{\mathrm{k,ins}} C^{\mathrm{ac/dc}}}{\eta^{\mathrm{ac}} \eta^{\mathrm{dc}}} \left(1 - \frac{2}{T^{\mathrm{ac/dc}}}\right)^T \times DF, \ \forall k \tag{22}$$

$$\upsilon_k^{\mathrm{su}} = \frac{p^{\mathrm{k,ins}} C^{\mathrm{su}}}{\eta^{\mathrm{dc}}} \left(1 - \frac{2}{T^{\mathrm{dc}}}\right)^T \times DF, \ \forall k \neq \mathtt{ta} \tag{23}$$

$$\upsilon_{bs}^{\mathrm{sw}} = C^{\mathrm{sw}} \left(1 - \frac{2}{T^{\mathrm{sw}}}\right)^T \times DF \tag{24}$$

$$\upsilon_{ta}^{\mathrm{oc}} = L^{\mathrm{ta}} C^{\mathrm{oc}} \left(1 - \frac{2}{T^{\mathrm{oc}}}\right)^T \times DF \tag{25}$$

$$\upsilon_k^{\mathrm{bat}} = \sum_{j=1}^{\lfloor T/t_k^{\mathrm{bat}} \rfloor} \left(\frac{v_{k,j}^{\mathrm{bat}}}{(1+int)^{\lfloor i \times t_k^{\mathrm{bat}} \rfloor}} \right) + v_{k,N}^{\mathrm{bat}} DF, \ \forall k \tag{26}$$

$$v_{k,j}^{\mathrm{bat}} = \omega_{k,j}^h W^u e_k^{\mathrm{cap}} C^{\mathrm{bat}} - e_k^{\mathrm{cap}} C^{\mathrm{rp/bat}}, \ \forall k \tag{27}$$

$$\omega_{k,j}^h = \frac{\kappa_{k,j}^b - EoL^b}{1 - EoL^b}, \ \forall k, j \tag{28}$$

$$\kappa_{k,j}^b = \begin{cases} 1 - e_k^{\mathrm{loss,yr}} (T - t_k^{\mathrm{bat}} \lfloor \frac{T}{t_k^{\mathrm{bat}}} \rfloor) & \forall j = N \\ EoL^{ev} & \text{Otherwise} \end{cases} \tag{29}$$

where $j \in \{1, 2, \ldots, N\}$ is the battery number index.

5. Simulation results and discussion

In this section, a hypothetical mine layout is used to evaluate the business value and performance of each charging system through an extensive scenario-based simulation study. In each scenario, the design parameters (e.g., truck battery size) are first obtained for each charging system following the procedures explained in Section 3. Then, a cost–benefit analysis is performed based on the design parameters obtained for each system.

All optimisation models were implemented in the General Algebraic Modelling System (GAMS). The fast-charging and battery-swapping formulations were solved as nonlinear programming problems using the global optimisation solver BARON, while the trolley-assist formulation was solved as a linear programming problem using the CPLEX solver. All problems were solved to optimality within solver tolerances.

5.1. Case study overview

The case study examines a typical underground mining haul profile, where a truck travels 500 m on flat terrain, then uphill with a 1:7 grade, and finally flat terrain to dump its payload, shown in Fig. 3. The truck returns empty downhill for reloading. Each shift is 12 h, with 9 h allocated to hauling.

The baseline truck used for comparison is the Sandvik TH663i diesel truck, which can carry up to 693 tonnes per shift. When fully loaded, it can go 8.6 km/h up a 1:7 grade slope, completing 11 trips per shift in 9 h of hauling time (considering a maximum safe speed of 15 km/h). However, if converted to a battery-powered electric truck capable of travelling uphill at 15 km/h, the travel time would decrease, allowing the electric truck to complete up to 14 trips per shift.

The loaded weight of the electric truck is assumed to be limited to 111 tonnes due to safety concerns, which is the fully loaded weight of the baseline diesel truck. Thus, if the battery and electric drivetrain weigh more than the diesel engine and full fuel tank (2300 kg), the payload capacity of the electric truck is reduced accordingly. According to the Sandvik TH550B [26], the 8260 kg battery can be swapped in 3 min, which is used as a standard measure to estimate battery exchange time in this paper. The weight of the motor, inverter, and battery are calculated from typical power/energy density ratings. Detailed specifications, component values for the electric truck and input parameters used in this study are provided in an electronic companion available at *Supplementary Material*.

5.2. Fast charging system

Assuming that a fast charging station is located at point A in Fig. 3, the design procedure explained in Section 3.1 is applied to the proposed haul profile. The uphill speed and the battery charging frequency are the two input parameters that change in each scenario. Without considering charging time, a truck can complete 11 trips at 8.6 km/h uphill (which is called 1× speed), Therefore, seven scenarios are simulated for the 1× speed, and eight scenarios are simulated for travelling at 15 km/h uphill (which is called 2× speed). For example, in the 3-trip-1× scenario, the truck stops to charge every 3 cycles while ascending the ramp at 1× speed. Table 1 presents the simulation results for four extreme scenarios that are selected from the total of 15 simulated scenarios. The first scenario, which involves charging the battery after each trip, is indicated by *Min*, while the scenario with the lowest charging frequency is represented by *Max*.

Based on the results, the Min-1× scenario requires the least battery capacity and installed charging power. However, the Min-2× scenario offers the best solution in terms of payload per shift due to the double uphill speed. Note that the onboard energy storage required (e^{str}) in both scenarios is the same, but the higher maximum drivetrain power is the decisive parameter to determine the battery size in the Min-2× scenario. Compared to Min-1× and Min-2× scenarios, Max-1× and Max-2× can have more trips per shift, as they avoid charging during a shift. However, their payload per trip is lower because they require a larger battery capacity. These scenarios use fewer battery cycles per year; hence, fewer battery replacements during the operation of the mine.

By applying the cost estimation model presented earlier to the four scenarios designed with fast charging system, the results shown in Table 2 are obtained. As can be seen, scenarios with 2× speed are more expensive than the corresponding scenarios with 1× speed, since they have a higher drivetrain power. In addition, scenarios with higher battery capacity require fewer charging stops, a higher battery capital cost, and a reduced number of battery replacements. The battery replacement cost in the Max-1× and Max-2× scenarios is zero because they can charge at a 0.5 C-rating and experience fewer cycles, both contributing to lower degradation and fewer replacements. Overall, the Min-1× scenario requires the minimum cost of implementation.

According to the AC2L index, the Min-1× scenario offers the lowest cost per tonne among the scenarios for the fast charging system.

[2] This shows the willingness of customers to pay for a used battery.

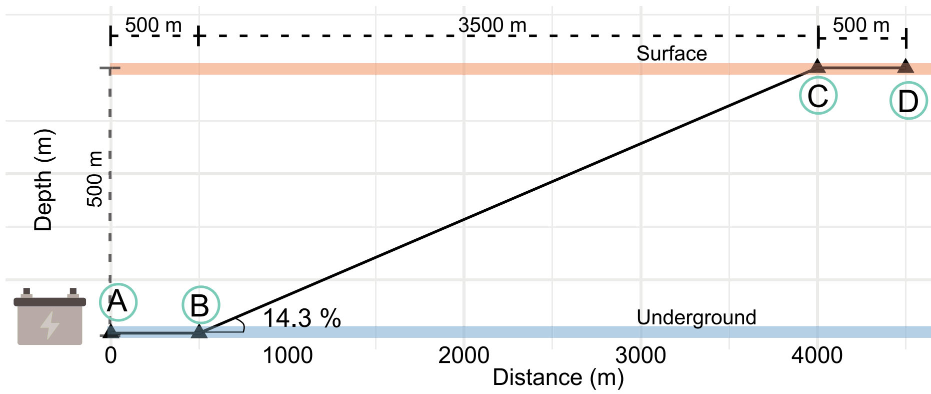

Fig. 3. Side view of the haul trip profile.

Table 1
Fast-charging and battery swapping analysis.

Trips Uphill speed	Min 1×	Min 2×	Max 1×	Max 2×
Battery capacity (kWh)	270	434	2247	2770
Drivetrain power (max, kW)	498	869	498	869
Weights (tonnes)				
Motor/Inverter	0.16	0.29	0.16	0.29
Battery	2.7	4.3	22.5	27.7
Vehicle (unloaded)	48.5	55.3	68.3	73.7
Payload (per trip)	62.4	60.7	42.7	37.3
Energy analysis (kWh)				
Required energy storage	216	216	1798	2216
Regenerative braking	42.2	43.7	59.4	64.1
FAST CHARGING:				
Battery charge time (hrs per trip)	0.32	0.19	–	–
Trip duration (h)	1.1	0.80	0.81	0.6
Trips per shift	8	11	11	14
Payload (tonnes/shift)	500	667	469	523
Payload (relative to diesel vehicle)	0.72	0.96	0.68	0.75
Power rating (MW)	0.54	0.87	1.1	1.4
Battery cycles per year (1000 s)	4.6	3.9	0.76	0.8
BATTERY SWAPPING				
Battery swap time (hrs per trip)	0.05	0.05	–	–
Trip duration (h)	0.83	0.65	0.78	0.6
Trips per shift	10	13	11	14
Payload (tonnes/shift)	624	789	469	523
Payload (relative to diesel vehicle)	0.9	1.14	0.68	0.75
Power rating (MW)	0.32	0.66	0.25	0.31
Battery cycles per year (1000 s)	2.9	2.3	0.38	0.39

Table 2
Fast charging system cost analysis.

Trips Uphill speed	Min 1×	Min 2×	Max 1×	Max 2×
Capital (M$)	0.62	0.936	1.822	2.222
Operation & Maintenance (M$)	1.063	1.519	1.466	1.804
Salvage value (M$)	0.216	0.360	0.648	0.799
NPV (M$)	1.467	2.095	2.640	3.227
AC2L metric ($/tonnes)	0.64	0.69	1.23	1.35

5.3. Battery swapping system

The battery swap station is assumed to be located at point Ⓐ of the haulage profile under study. Using the design framework introduced in Section 3.1, the simulation results are obtained and presented in Table 1. In a similar manner to the fast charging system, seven scenarios are simulated at 1× speed, while eight scenarios are simulated at 2× speed. Compared to the fast charging system, two batteries are required for each scenario. However, the swapping time is less than the charging time and consequently an electric truck can have more trips per shift.

Table 3
Battery swapping system cost analysis.

Trips Uphill speed	Min 1×	Min 2×	Max 1×	Max 2×
Capital (M$)	0.836	1.223	2.281	2.723
Operation & Maintenance (M$)	1.205	1.632	1.629	1.981
Salvage value (M$)	0.304	0.482	1.04	1.304
NPV (M$)	1.736	2.373	2.870	3.439
AC2L metric	0.61	0.66	1.34	1.44

In this way, the Min-2× scenario exceeds the baseline payload per shift. Since batteries have extended periods available for recharging, the charging power needed is reduced. In addition, two battery packs are used in a shift, which means that each pack experiences fewer cycles, leading to lower battery replacement costs.

Table 3 shows the results of the cost analysis of the battery swapping system. It is evident that the Min-1× scenario is the best solution in terms of the AC2L index, while the Min-2× scenario is capable of transporting more material in a shift. By comparing the results for the battery swapping system and the fast charging system, it can be seen that the Min-1× scenario of the battery swapping system is the best solution in terms of the AC2L index. In addition, it can be seen that the Max-1× and Max-2× scenarios demonstrate better performance under fast charging compared to battery swapping. This is because, in these scenarios, the fast charging system results in reduced costs while preserving the payload capacity identical to that of battery swapping.

5.4. Trolley assist system

As mentioned earlier, it is assumed that the trolley line starts at the point Ⓒ in Fig. 3 and extends down the slope. The uphill speed and length of the overhead trolley line are the two input parameters that change in each scenario. At each uphill speed, ten scenarios are simulated, each featuring a 10% change in the length of the trolley line. Implementing the design procedure presented in Section 3.3 in Pyomo and solving the optimisation problem using `glpk` solver, the results shown in Table 4 are obtained. In scenarios where the entire ramp is serviced by the overhead trolley line, the required battery capacity is considerably reduced, allowing for a greater payload capacity. Although scenarios with higher uphill speeds have less payload capacity due to the larger battery, they have a larger payload per shift than comparable scenarios operating at lower uphill speeds because of the higher number of trips they can achieve per shift.

The cost analysis of the trolley assist system is presented in Table 5. As can be seen, scenarios with a longer trolley line need less initial battery capacity, yet have a high cost of trolley line construction, which is their dominant cost component. Scenarios with trolley lines that cover half the incline length lead to significantly reduced expenses,

Table 4
Overhead trolley analysis.

Length of incline with trolley lines Uphill speed	100% 1×	100% 2×	50% 1×	50% 2×
Battery capacity (kWh)	86.9	278	249.1	465.6
Battery power (max discharge, kW)	151.2	151.2	498.2	869
Drivetrain power (max, kW)	498.2	869	498.2	869
Weights (tonnes)				
Motor/Inverter	0.16	0.29	0.16	0.29
Battery	0.87	2.78	2.49	4.65
Vehicle (unloaded)	46.7	48.75	48.34	50.62
Payload	64.3	62.25	62.66	60.37
Energy analysis (kWh)				
Uphill supply energy	178.2	88.3	139.4	87.55
Uphill battery discharging	26.6	116.5	−36.97	14.86
Downhill charging	0	88.3	37.52	87.55
Regenerative braking	40.97	42.75	42.39	44.4
Trip duration (h)	0.78	0.6	0.78	0.6
Trips per shift	11	14	11	14
Payload (tonnes/shift)	707	871	689	845
Payload (relative to diesel vehicle)	1.02	1.26	0.99	1.22
Power rating (MW)	0.43	0.37	0.68	0.74
Battery cycles per year (1000 s)	2.46	4.28	1.21	1.92

Table 5
Trolley assist system cost analysis.

Length of incline with trolley lines Uphill speed	100% 1×	100% 2×	50% 1×	50% 2×
Capital (M$)	3.214	3.272	1.854	1.953
Operation & Maintenance (M$)	2.208	2.506	1.651	1.993
Salvage value (M$)	0.615	0.661	0.385	0.459
NPV (M$)	4.807	5.117	3.119	3.486
AC2L metric	1.49	1.28	0.99	0.90

while their payload capacity is not as much as compared to the overall cost. In this context, the two scenarios that use a 50% trolley line exhibit significantly improved AC2L values. In particular, the 50%-2× scenario demonstrates superior performance with respect to the AC2L metric, as it is capable of moving a larger payload.

5.5. Further discussions

To compare the performance of the three charging systems in different scenarios, Fig. 4 illustrates the NPV of each charging system against payload per shift. In the context of fast charging and battery swapping, the numbers inside each symbol in Fig. 4 indicate the number of trips possible without requiring a recharging stop, based on the battery size designed for that scenario. For the trolley charging system, the percentages within each square reflect the portion of the slope served by the overhead trolley line. The horizontal dashed line in the figure represents the payload performance of the baseline diesel truck. Scenarios where the number of planned trips exceeds half of the total required trips without necessitating a recharge during a shift have been excluded, due to their relatively poorer performance.

It is evident that the 1-trip-1× (scenario 1 with 1× speed) scenario under the fast charging system represents the most cost-effective option; however, it significantly fails to reach the baseline payload performance. Among fast charging and battery swapping systems, the 2-trip-2× scenario for the battery swapping system exhibits superior payload capacity performance. Also, it is interesting to observe that the 11-trip-1× scenario in the fast charging system incurs a lower NPV compared to the 4-trip-1×, 5-trip-1×, and 6-trip-1× scenarios. This is because the 11-trip-1× scenario allows more time for recharging the depleted battery, which in turn allows less expensive and lower power charging equipment. This pattern is also evident in scenarios involving the fast charging system based on 2× speed configurations. However, since the cost of batteries is the primary factor in the battery swapping system due to the need for a spare battery, this trend does not occur in these systems.

In the case of the trolley assist charging, the cost for overhead lines is the dominant cost for almost all scenarios. Increasing the length of the trolley line increases the NPV in most cases, as shown in Fig. 4. In addition, the power required for the electric truck drivetrain in the uphill trip is the decisive factor for the required battery capacity and the power level of the trolley in some of the scenarios. Therefore, the designed battery size and drivetrain power are the same, and hence the battery and motor/inverter weights and consequently the payload capability for those scenarios are equal.

Furthermore, as demonstrated in Fig. 4, the NPV associated with the trolley assist system for the 10%-1× scenario is greater than the 20%-1× scenario, despite the latter having a trolley line length that is twice as long. This is attributed to the need for the 10% scenario to use a high power level for fast battery charging, which requires a large battery capacity capable of handling such power due to the C-rating limit. This also happens for the 10%-2× and 20%-2× scenarios.

Fig. 5 also shows the AC2L values for different design scenarios under three charging systems. As can be seen, the 1-trip-1×, 2-trip-1×, and 2-trip-2× for the battery swapping system have the best AC2L values among all the design scenarios for fast charging and battery swapping systems. In addition, the 1-trip-1× scenario has the best AC2L value in the fast charging system scenarios. Although trolley assist system exhibits excellent performance with respect to payload capacity, it is less appealing due to its significant capital expenditure. However, among the different scenarios for the trolley charging system, the 20%-1×, 20%-2×, and 30%-2× scenarios have a competitive AC2L value compared to the other systems. Moreover, in trolley assist charging system, 2× scenarios generally outperform those with 1× in terms of AC2L values. This is attributed to their improved payload capability, while NPV remains more stable relative to payload capability for these cases, as illustrated in Fig. 4.

6. Conclusions and future work

This paper presents a comprehensive scenario-based framework for evaluating charging system options for electrified mining haul trucks, offering a structured approach to quantify the trade-offs between charging technologies, battery capacities, and operational performance. The framework incorporates detailed technical models for fast charging, battery swapping, and trolley assist solutions, and integrates key factors such as battery degradation, charging power requirements, and system-level operational constraints within a net present cost evaluation.

To demonstrate the application of the proposed framework, a case study of an underground mining operation was analysed. The results illustrate how different charging scenarios impact both the cost and productivity metrics. Battery swapping scenarios achieved favourable annualised cost per tonne performance. The trolley assist configurations offer a higher payload capacity but require greater capital investment. Fast charging solutions presented lower initial costs, but resulted in reduced productivity due to charging delays. Although these findings are specific to the analysed mine profile, they illustrate the ability of the proposed framework to capture the interdependencies between charging-system type and sizing, battery mass, cycle-time dynamics, and long-term economic impacts. As such, the proposed methodology provides mining operators with a transparent tool for comparing electrification pathways based on their operational characteristics, cost constraints, and productivity objectives.

Future research directions include investigating the potential benefits of combining multiple charging systems in a single mining operation. This will require the development of multi-objective optimisation models capable of evaluating mixed charging architectures across diverse operating conditions. Furthermore, future work should examine the integration of mine site energy production systems with the design of the charging system, including the potential role of the onboard battery as a distributed storage resource, potentially unlocking additional value from the battery capacity of haul trucks.

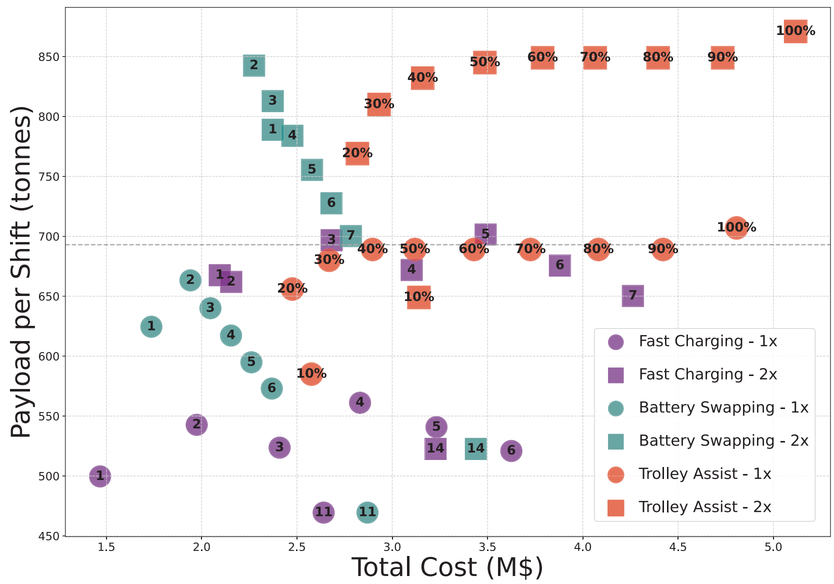

Fig. 4. NPV versus payload capability of different charging systems.

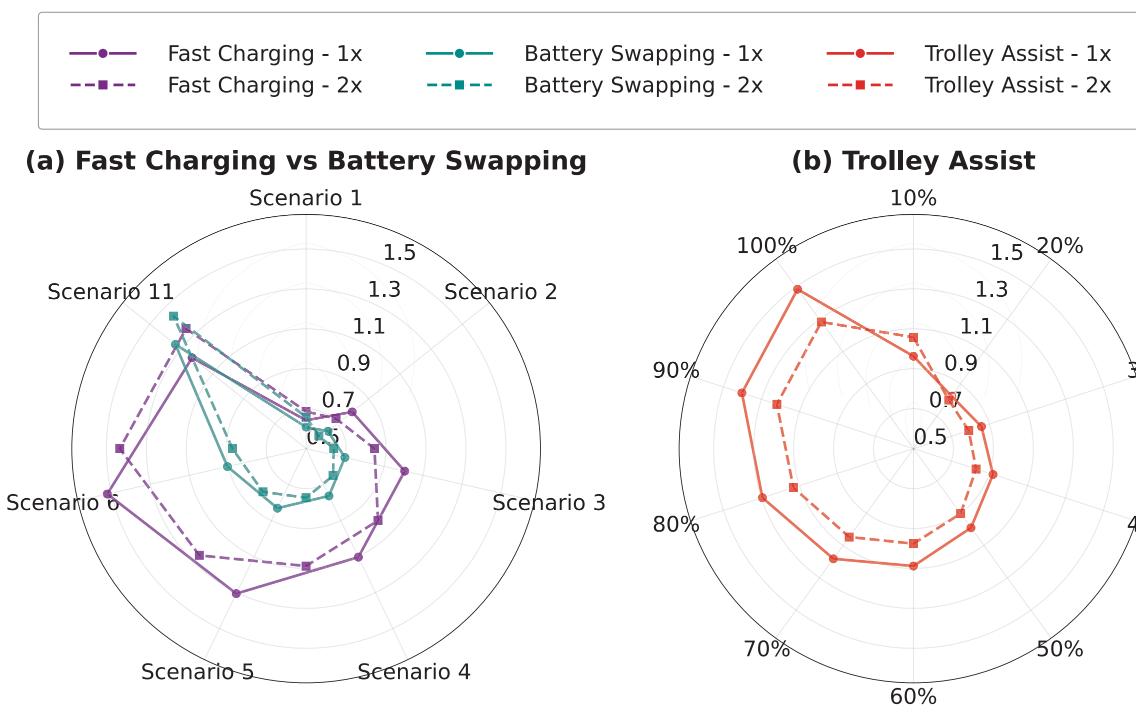

Fig. 5. AC2L values for different charging systems.

CRediT authorship contribution statement

Seyed Nasrollah Hashemian: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Hirad Assimi:** Writing – review & editing, Visualization, Supervision, Software, Data curation. **Hossein Ranjbar:** Writing – review & editing, Visualization, Validation. **S. Ali Pourmousavi:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Wen L. Soong:** Writing – review & editing, Validation, Supervision, Formal analysis, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Writefull for grammar checking. After using this tool the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All data generated or analysed during this study are included in this published article.

References

[1] SunSHIFT. Renewable energy in the Australian mining sector: White paper. Tech. rep., Aust. Renewable Energy Agency (ARENA); 2017.
[2] Assimi H, Soong W, Pourmousavi Kani A, Hashemian SN, Islam S, Foley B. An overview of Australia's mining vehicle and equipment electrification. Tech. rep., Future Battery Industries Coop. Res. Centre (FBICRC) Ltd; 2023.
[3] Assimi H, Ataabadi SNH, Islam SMM, Soong WL, Pourmousavi SA. Toward underground mobile fleet electrification: Three essential steps to make a real change. IEEE Electr Mag 2024;12(1):16–26.
[4] Tokac B, Zhang Q, Sari YA. Environmental and economic comparison of diesel and electric trucks in open-pit mining operations. J Clean Prod 2025;145540.
[5] Ertugrul N, Pourmousavi Kani A, Davies M, Sbarbaro D, Morán L. Status of mine electrification and future potentials. In: 2020 int. conference on smart grids and energy systems. SGES, 2020, p. 151–6. http://dx.doi.org/10.1109/SGES51519.2020.00034.
[6] Zhang H, Qiu J, Wang Y. Planning strategy of fast-charging stations in coupled transportation and distribution systems considering human health impact. Int J Electr Power Energy Syst 2021;133:107316. http://dx.doi.org/10.1016/j.ijepes.2021.107316.
[7] Wu C, Wang Y, Shi Q, Gao S. A two-layer planning method for location and capacity determination of public electric vehicle charging stations. Int J Electr Power Energy Syst 2024;161:110205. http://dx.doi.org/10.1016/j.ijepes.2024.110205.
[8] Acharya S, Mieth R, Karri R, Dvorkin Y. False data injection attacks on data markets for electric vehicle charging stations. Adv Appl Energy 2022;7:100098. http://dx.doi.org/10.1016/j.adapen.2022.100098.
[9] Perumal SS, Lusby RM, Larsen J. Electric bus planning & scheduling: A review of related problems and methodologies. European J Oper Res 2021. http://dx.doi.org/10.1016/j.ejor.2021.10.058.
[10] Chen Z, Yin Y, Song Z. A cost-competitiveness analysis of charging infrastructure for electric bus operations. Transp Res C 2018;93:351–66.
[11] Ahmad Iqbal M, I. Almaraj I. Short-term scheduling optimization of battery electric buses in the context of sustainable energy resources under uncertainty. Int J Electr Power Energy Syst 2025;168:110715. http://dx.doi.org/10.1016/j.ijepes.2025.110715, URL https://www.sciencedirect.com/science/article/pii/S0142061525002662.
[12] Yıldırım Ş, Yıldız B. Electric bus fleet composition and scheduling. Transp Res C 2021;129:103197.
[13] Mareev I, Becker J, Sauer DU. Battery dimensioning and life cycle costs analysis for a heavy-duty truck considering the requirements of long-haul transportation. Energies 2017;11(1):55.
[14] Sandvik Mining and Rock Solutions. TH665b battery electric truck success at Sunrise Dam, Australia. 2024, URL https://www.youtube.com/watch?v=KZf2zw82tS4, [Accessed 23 December 2024].
[15] Ratcliffe M, Mahoney P, Stanway G. State of play: Electrification. 2020, URL https://bit.ly/3zyjbrE, [Accessed 7 January 2025].
[16] The electric mine consortium: a case study in transformative collaboration. 2022, URL https://electricmine.com, [Accessed 7 January 2025].
[17] Electric Mine Simulation Challenge. 2021, URL https://unearthed.solutions/u/competitions/electric-mine-simulation, [Accessed 7 January 2025].
[18] Rafi MAH, Rennie R, Larsen J, Bauman J. Investigation of fast charging and battery swapping options for electric haul trucks in underground mines. In: 2020 IEEE transp. electrification conf. & expo. ITEC, IEEE; 2020, p. 1081–7.
[19] Meshginqalam A, Mobarak MH, Dong J, Rafi MAH, Rennie R, Bauman J. An optimization algorithm for the design of battery electric fleets in underground mines. IEEE Access 2024.
[20] Bao H, Knights P, Kizil M, Nehring M. Performance evaluation of alternative configurations of battery electric mining haul trucks. IEEE Access 2025.
[21] Valenzuela Cruzat J, Valenzuela MA. Modeling and evaluation of benefits of trolley assist system for mining trucks. IEEE Trans Ind Appl 2018;54(4):3971–81. http://dx.doi.org/10.1109/TIA.2018.2823261.
[22] Mareev I, Sauer DU. Energy consumption and life cycle costs of overhead catenary heavy-duty trucks for long-haul transportation. Energies 2018;11(12):3446.
[23] Bao H, Knights P, Kizil M, Nehring M. Energy consumption and battery size of battery trolley electric trucks in surface mines. Energies 2024;17(6):1494.
[24] Wei L, Liu Q, Liu B, Feng Y. Optimal economic sizing of energy storage system for trolley assisted battery electric mining haul truck considering battery degradation. J Clean Prod 2025;524:146452. http://dx.doi.org/10.1016/j.jclepro.2025.146452, URL https://www.sciencedirect.com/science/article/pii/S0959652625018025.
[25] Ehsani M, Gao Y, Longo S, Ebrahimi K. Modern electric, hybrid electric, and fuel cell vehicles. CRC Press; 2018.
[26] Sandvik TH550B haul truck specifications. 2021, URL https://bit.ly/3ztOe7Q, [Accessed 7 January 2025].
[27] Francfort J, Garetson T, Karner D, Salisbury SD, Smart JG. Considerations for corridor and community DC fast charging complex system design. Tech. rep., Idaho National Lab.; 2017.
[28] Wang J, Purewal J, Liu P, Hicks-Garner J, Soukazian S, Sherman E, Sorenson A, Vu L, Tataria H, Verbrugge MW. Degradation of lithium ion batteries employing graphite negatives and nickel–cobalt–manganese oxide+ spinel manganese oxide positives: Part 1, Aging mechanisms and life estimation. J Power Sources 2014;269:937–48.
[29] Neubauer J, Pesaran A. The ability of battery second use strategies to impact plug-in electric vehicle prices and serve utility energy storage applications. J Power Sources 2011;196(23):10351–8.